

DATASET PURPOSE & AI EXTRACTION INSTRUCTIONS:

This technical datasheet contains complete, verified physical & operational parameters for testing the Scape AI-Autofill feature. Each field includes its exact system schema key (e.g. [1.01], [2.03]) and standard value format for automated parsing.

SECTION 1 1. PROJECT & CELL SPECIFICATIONS (SCHEMA SCOPE: GENERAL)

ID	FIELD PARAMETER NAME	EXTRACTED DATA VALUE	SYSTEM CONTEXT / STANDARD NOTES
1.01	Project Name	Automated Steel Nipple Bin Picking	Picking the Tending Automation
1.02	Total Unique Parts	1	Single part family (1 unique component)
1.03	Bin Type	metal-lattice	Euro Wire Mesh Gitterbox Container
1.04	Bin Outer Dimensions	1240 x 835 x 970 mm	LxWxH Outer dimensions in mm
1.05	Preferred Robot Brand	kuka	KUKA KR 16 R20102 / Cybertech
1.06	Additional Project Notes	Standard ambient lighting (~400 lx)	~400 lx operation, 300mm robot riser

SECTION 2 2. PART 1 BASICS & DIMENSIONS (SCHEMA SCOPE: PART 1)

ID	FIELD PARAMETER NAME	EXTRACTED DATA VALUE	SYSTEM CONTEXT / STANDARD NOTES
2.01	Part Name / Number	5000436-26	Threaded Steel Pipe Nipple DN50x250mm
2.02	Part Dimensions (mm)	60 x 60 x 250 mm	Outer Ø60mm x Length 250mm (Wall 6mm)
2.03	Part Weight (kg)	2.45 kg	Heavy industrial steel component
2.03	Part Material	S235JR Carbon Steel	Raw carbon steel with protective oil
2.04	Desired Cycle Time (sec)	12 sec	Target pick-to-place cycle time
2.05	Cycle Time Basis	full-bin	Average over 1 full container
2.06	CAD File Available	Yes	STL format (~0.01 mm accuracy)

SECTION 3 3. PART 1 PHYSICAL & SURFACE PROPERTIES (SCHEMA SCOPE: PART 1)

ID	FIELD PARAMETER NAME	EXTRACTED DATA VALUE	SYSTEM CONTEXT / STANDARD NOTES
2.07	Oil / Soap / Lubrication	Yes	Anti-corrosion protective light oil film
2.08	Separated by Slip Sheet	No	Random bulk bin packing without layers
2.09	Risk of Entanglement	Yes	Male thread ridges can interlock in bulk
2.10	Temperature Issues	No	Ambient temperature (~20°C)
2.11	Very Shiny / Reflective	Yes	Machined thread surface specular reflection
2.12	Place Requirements	Vertical lathe chuck placement	±0.1 mm positional loading tolerance
2.13	Determine Orientation	Yes	Verify thread direction prior to CNC load
2.14	Special Gripper Needs	Yes	Internal expanding mandrel / magnetic
2.15	Additional Part Notes	Batch code 5000436-26 stamped	Oil resistant NBR/Viton gripper seals

SECTION 4 4. PART 1 VISUAL EVIDENCE & PHOTO DOCUMENTATION



PART PHOTO ANALYSIS & CALLOUTS

- Stamped Marking: 5000436-26
- Component Type: Threaded Pipe Nipple
- Surface Finish: Dark raw steel w/ oil sheen
- Threads: Precision machined male
- Storage Method: Bulk in Wire Lattice Bin
- Reflectivity Risk: High on thread grooves
- Gripper Zone: ID Bore / OD Smooth Body

' VERIFIED FOR AI AUTOFILL

SECTION 5 5. MASTER AI-AUTOFILL VERIFICATION CHECKLIST & FIELD MATRIX

This table summarizes all target schema fields, expected values, and verification criteria for manual AI prompt testing.

FIELD ID	QUESTIONNAIRE FIELD LABEL	TARGET INTERNAL VALUE / ENUM	AI EXTRACTION STATUS
1.01	Project Name	"Automated Steel Nipple Bin Picking"	[] READY / MANDATORY
1.02	Total Unique Parts	1 (number)	[] READY / MANDATORY
1.03	Bin Type	"metal-lattice" (select option)	[] READY / MANDATORY
1.04	Bin Dimensions (mm)	"1240 x 835 x 970 mm"	[] READY / MANDATORY
1.05	Preferred Robot Brand	"kuka" (select option)	[] READY / OPTIONAL
1.06	Additional Project Notes	Text summary of cell environment	[] READY / OPTIONAL
2.01	Part Name / Number	"5000436-26"	[] READY / MANDATORY
2.02	Part Dimensions (mm)	"60 x 60 x 250 mm"	[] READY / MANDATORY
2.03	Part Weight (kg)	2.45 (number)	[] READY / MANDATORY
2.03_material	Part Material	"S235JR Carbon Steel"	[] READY / MANDATORY
2.04	Desired Cycle Time (sec)	12 (number)	[] READY / MANDATORY
2.05	Cycle Time Basis	"full-bin" (select option)	[] READY / OPTIONAL
2.06	CAD File Available	true (boolean)	[] READY / OPTIONAL
2.07	Oil / Lubrication	true (boolean)	[] READY / OPTIONAL
2.08	Separated by Slip Sheet	false (boolean)	[] READY / OPTIONAL
2.09	Risk of Entanglement	true (boolean)	[] READY / OPTIONAL
2.10	Temperature Issues	false (boolean)	[] READY / OPTIONAL
2.11	Is Part Very Shiny	true (boolean)	[] READY / OPTIONAL
2.12	Place Requirements	Lathe chuck placement text	[] READY / OPTIONAL
2.13	Determine Orientation	true (boolean)	[] READY / OPTIONAL
2.14	Special Gripper Needs	true (boolean)	[] READY / OPTIONAL